

Capstone Project: Multi-Agent AI Platform with GenAI, Analytics & Azure Deployment

Left Shift Program 2026 – Data & AI (T5)

Type: Individual Project

Complexity: Intermediate → Expert

1. Project Overview

Students will design, build, and deploy an **end-to-end Multi-Agent AI Platform** that integrates:

- **Python Full-stack development**
- **Machine Learning / Deep Learning**
- **GenAI (LLMs, RAG, embeddings, multi-agent orchestration)**
- **Azure AI services**
- **Azure Data Engineering & Analytics stack**
- **Power BI visualization**
- **Azure deployment / CI-CD fundamentals**

The project simulates a real-world Data + AI engineering scenario, requiring the student to integrate skills learned across the full track.

2. Problem Statement (Choose ONE of the following domains)

Students must pick one domain and build their end-to-end solution around it:

1. **Smart Retail Assistant** – Demand forecasting + customer Q&A + anomaly detection
 2. **Intelligent Healthcare Support System** – Symptom triage + medical information assistant + patient data analytics
 3. **Financial Risk & Advisory Platform** – Fraud detection + portfolio insights + regulatory search assistant
 4. **Manufacturing Quality & Productivity Suite** – Defect classification + production optimization + maintenance insights
 5. **Custom domain scenario (with SME consent)**
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3. Mandatory Components

A. Python Fullstack (Backend + APIs)

Students must create:

- Python-based backend using **Flask or FastAPI**

- Minimum **4 REST APIs** (data ingestion, ML prediction, document search, agent interaction)
 - Integration with any one database:
 - **SQL** (Azure SQL) or
 - **NoSQL** (MongoDB, CosmosDB)
 - Basic **logging, error handling, and unit testing (pytest)**
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B. Machine Learning / Deep Learning

Implement at least **one ML/DL model**, e.g.:

- Regression / classification (sklearn)
- Clustering
- Time-series forecasting
- Image classification (CNN / OpenCV)

Deliverables:

- Clean data pipeline
 - Feature engineering
 - Training + evaluation
 - Model persistence (pickle / ONNX)
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C. GenAI / Agents

Build a **minimum 2–3 agent system** using:

- Prompt engineering
- Embeddings + vector store
- Retrieval-Augmented Generation (RAG)
- Agents (Autogen / LangChain / CrewAI / Azure GenAI)

Examples:

- **Data Analyst Agent** – answers questions from analytics data
- **Document Assistant Agent** – searches PDFs / knowledge base
- **ML Expert Agent** – helps generate insights from ML outputs

Implement:

- Multi-agent orchestration

- Agent communication through Model Context Protocol (MCP) *optional but recommended*
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D. Azure AI & Cloud

Use at least **three Azure components**:

- **Azure OpenAI / Azure AI Studio**
- **Azure Bot Service** or Web App
- **Azure Cognitive Services** (Search, Vision, Document Intelligence, Text Analytics)
- **Azure ML** for training or inference
- **Azure AI Foundry** for deployment patterns

Include:

- Deployment diagram
 - Security consideration (Key Vault, environment variables)
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E. Data Engineering Pipeline

Using any combination of:

- **Azure Data Factory** – ingest raw data
- **Azure Databricks (PySpark)** – clean, transform, analyze data
- **Azure Fabric** – Lakehouse, pipelines, Data Activator
- **SQL** (Spark SQL or T-SQL)

Deliver:

- Raw → Staged → Curated data flow
 - Delta tables or parquet-based storage
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F. Analytics & Visualization

Build a **Power BI dashboard** showing:

- Key metrics
- Model outputs
- Anomaly alerts / trends
- Agent-driven insights (optional)
- Publish & share report

G. Final Deployment

Deploy the solution on Azure using one of:

- **Azure Web App / Container App**
- **Docker + GitHub Actions CI/CD**
- **Azure App Service**
- **Serverless Functions (for tasks)**

Your multi-agent system must be **live / executable** during evaluation.

4. Deliverables

Deliverable	Description
Technical Documentation	Architecture, data flow, models, prompts, agents, APIs
Working Code Repository	GitHub link with clear folder structure
Deployment Diagram	Azure components + interactions
Configuration Files	Environment, pipeline definitions, YAML, etc.
Power BI Report	Final analytics dashboard
Demo Video (5–10 mins)	Walkthrough of platform
Reflection Note	Challenges + learnings + optimizations

5. Evaluation Criteria

Design & Architecture
Python Full stack & APIs
ML/DL Model Quality
GenAI & Multi-Agent Implementation
Azure Deployment & Data Engineering
Documentation & Presentation

6. Optional Bonus Features

- MCP-based agent interoperability
- Real-time streaming analytics (Event Hub + Data Activator)
- LLM fine-tuning or custom embeddings
- Automated CI/CD with GitHub Actions